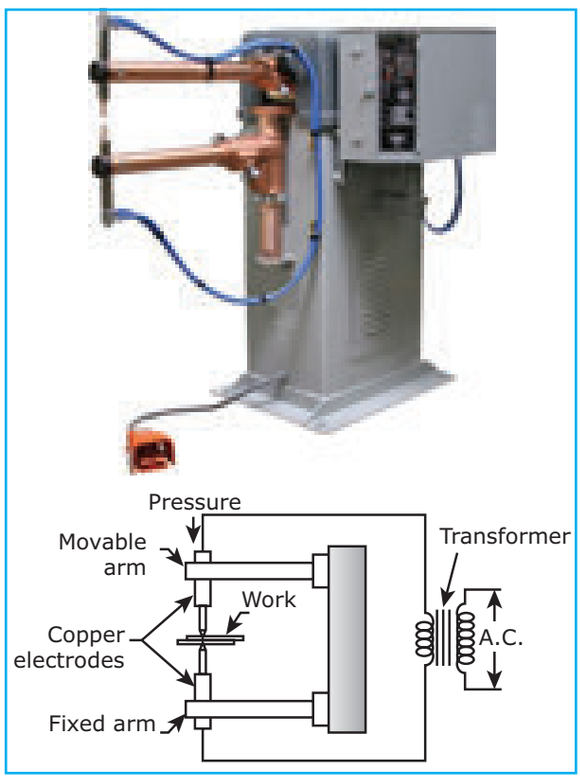
The strength of the joint is more	The joint is not as strong as that of arc welding
Brazing and soldering cannot be performed by the electric arc.	The processes of brazing and soldering can be done with the gas flame.
The filler rod metal should be selected as the same metal as that of the parts of the joint.	The filler rod metal can be different from that of the parts of the joint.

7.17 Resistance welding

The process of resistance welding involves

Developing electrical resistance in the parts of the joint to bring them into a plastic state and

Applying pressure on the parts to make the joint.



Resistance welding

Two copper electrodes are connected to an electric circuit of low resistance. The parts to be welded are placed between the electrodes. When current is allowed to pass through the electrodes, high electrical resistance is developed at the joint. Because of the resistance, heat is generated at the joint. The metal parts reach plastic state at this high temperature.

At this point, pressure is applied by means of either mechanical or hydraulic or pneumatic power source to make the joint. Current is provided by a suitable A.C. transformer. Resistance welding is useful in welding sheet metal, bars and pipes.

7.18 Welding related processes

Following are some important welding related processes used in industries.

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1. Oxy-acetylene cutting
2. Arc cutting
3. Hard facing
4. Soldering
5. Brazing

Oxy-acetylene cutting

Plates made of iron and steel are cut by oxygen, acetylene cutting torch. Oxygen and acetylene are mixed in the cutting torch and a gas flame is generated. The flame heats the sheets to bring them to red hot condition. High pressure oxygen is supplied on the red hot metal through a separate central hole of the cutting torch. Iron and steel sheets are thus cut by the jet of oxygen.

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